PAPERWEIGHT

PROJECT NAME:	Custom Flight Controller {An Attempt with Learning}
PROJECT GROUP:	Sahil Khandare & Pratik Khadse

WHY THIS PROJECT ?

- This project was undertaken to create an alternative Flight Controller. Our goal is to design and develop a Custom Flight Controller specifically to our requirements. These requirements will evolve based on the future competitions the club participates in.
- Our mission is to develop and build all electronic components in-house, ensuring full control over performance, customization and innovation.

INTRODUCTION

- This project was developed using an STM microcontroller integrated with sensors such as an IMU and a barometer. The flight controller PCB was designed and fabricated entirely in-house, and ChibiOS was selected for firmware development. However, during firmware flashing, the board failed due to the absence of proper ESD protection, which led to damage to the microcontroller and prevented successful programming.

KEY FEATURES {WE THOUGHT}

1. ChibiOS-Based Firmware Architecture
   The firmware was planned to be developed using ChibiOS on the SpeedyBee F405 .
2. In-House PCB Development:
   Custom-designed PCB tailored specifically for the future expansion.
3. Sensor Integration:
   The controller was designed to interface with an IMU and barometric sensor to provide accurate attitude estimation, altitude measurement, and stable flight control.

ARCHITECTURE

1. Microcontroller Integration:
   STM32F405RGT6
2. Barometer Selection:
   BMP200
3. IMU Selection :
   ICM 42670

RESULTS

- The project provided valuable hands-on learning in custom flight controller design, including hardware integration, PCB development, and real-time firmware planning. Although firmware flashing failed due to the absence of ESD protection, this highlighted the critical importance of robust hardware protection, proper grounding, and design-for-reliability practices. Overall, the experience strengthened understanding of embedded system design challenges and informed improvements for future flight controller revisions.

FUTURE SCOPE

1. **Another Attempt**